# Introduction to Digital Circuits

Lecture Notes

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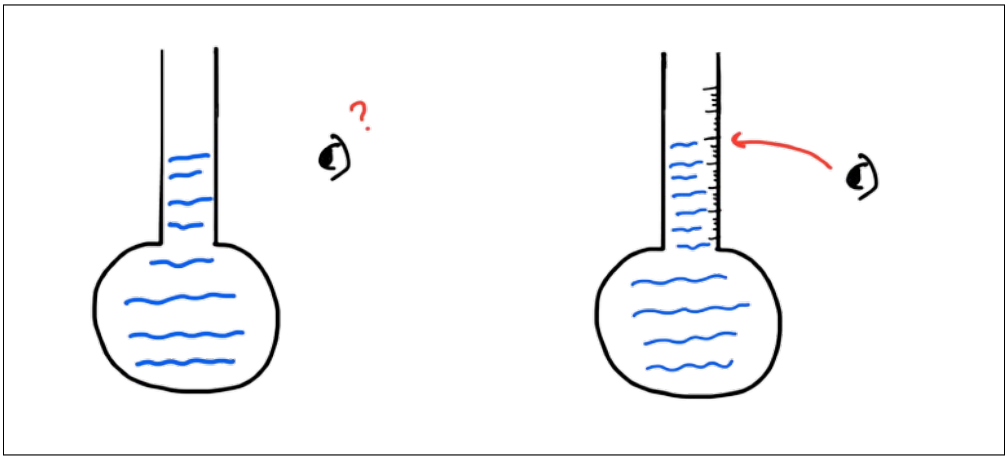
# Introduction

We've explored analog components like **diodes**, **Op-Amps**, and **BJTs**. These form the foundation of analog electronics — where signals vary continuously.

With the discovery of electricity, devices like the *telegraph* used *Morse Code* to send signals faster than ever. Electricity gave speed and scale to communication.

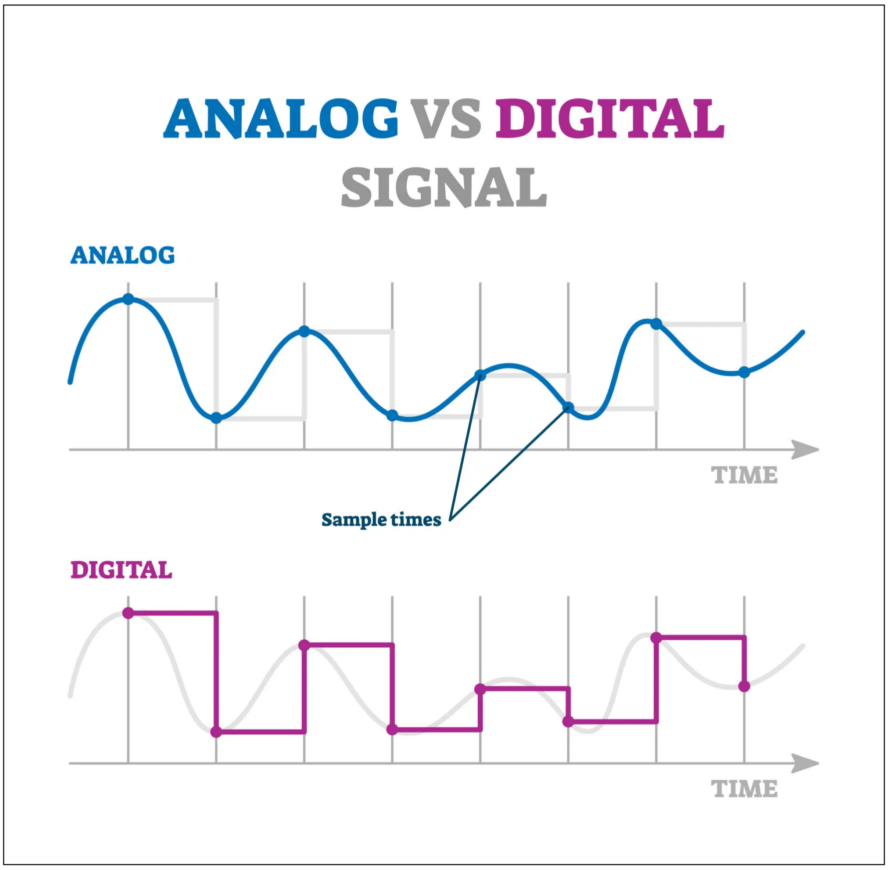
Now, digital systems allow precision and consistency. Imagine a thermometer with no labels — each person reads it differently. But if you label it numerically, everyone gets the same value. That's digital logic in action!




## Analog vs. Digital Signals

Table 1: Comparison between Analog and Digital Signals

| **Analog Signals** | **Digital Signals** |
|---|---|
| Continuous and time-varying | Discrete in time |
| Can take any value within a range | Limited to defined values (typically 0 and 1) |
| Susceptible to noise and distortion | More resistant to noise; easier to store and transmit |
| Example: Microphone output, temperature sensor | Example: Digital clock, computer processor data |




### Signal Conversion

Modern devices use both analog and digital signals.

- **ADC:** Converts analog voltage to binary data.
- **DAC:** Converts digital binary back to analog signal.

Used in audio, sensors, microcontrollers, and control systems.

## Digital Representation of Information

### What is Digital Representation?

Digital systems use 0 and 1 to represent all information. Switches, BJTs, and capacitors physically store these states.

A switch (ON/OFF) is all you need to think binary!

### Precision vs. Depth

Digital signals offer consistent results despite noise — but limit how detailed a signal can be.

- More precision → cleaner interpretation.
- Fewer states → less detail (depth).

**Example**

An Op-Amp reading 3.6V or 4.2V would both be interpreted as 5V digitally (if >2.5V). The fine detail is lost, but classification is preserved.

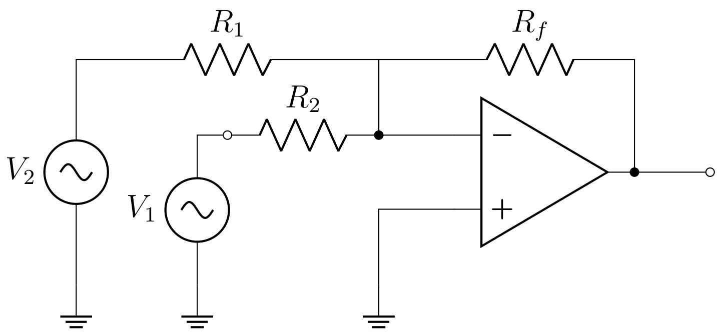



Figure 1: Summing Amplifier Op-Amp circuit

## Analog Signals in the Real World

All real-world signals — like sound, light, and temperature — are **analog**. To store or process them digitally:

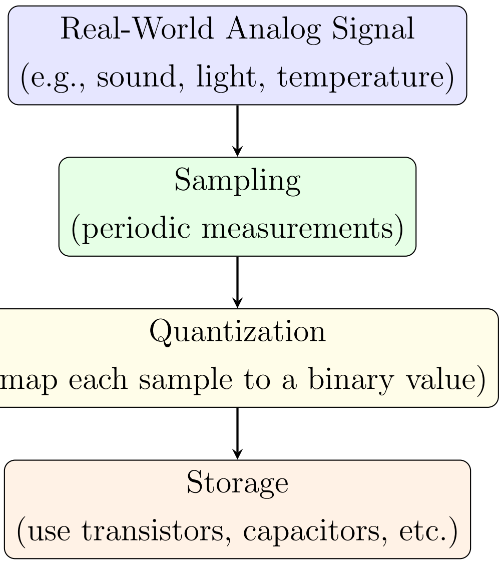



Figure 2: Flowchart for converting Analog Signals to Digital Form

**Example**

A microphone converts sound into voltage. This voltage is sampled and stored as binary numbers (e.g., 01001011), which can later be played back.

## Bits, Bytes, and States

Every switch (bit) can be either 0 or 1, giving:

$$\text{Number of states} = 2^n \quad \text{for } n \text{ bits}$$

**Example**

If you want 0.1V precision over 0–5V:

$$\frac{5}{0.1} = 50 \text{ states} \Rightarrow n = \lceil \log_2 50 \rceil = 6$$

Thus, 6 switches (bits) are needed.

**More bits → More states → More depth**

$$\textbf{8 bits = 1 byte} \Rightarrow 2^8 = 256 \textbf{ states}$$

Each byte can represent a character (e.g., ASCII), a pixel in an image, or a sound amplitude.

## How Do Computers Understand Data?

Everything in a computer — text, images, music — is stored as bits. For example:

$$\texttt{01000011} = \text{`C' (ASCII code 67)}$$

Pixels in images = 1 byte per color channel. Every sound sample = 1 byte or more. It's all bits!

### Physical Representations of Bits

- Capacitor → holds charge = 1, no charge = 0
- Magnetic bit → direction of magnetic field

- SSD → floating-gate transistors
- CD → tiny pits and lands
- Punch Card → holes represent data

Today, a USB the size of a coin can store **2 TB = 16 × $10^{12}$ bits**!

**Did You Know?**

Morse Code is a binary system! Each character is represented using dots (short pulse) and dashes (long pulse) — similar to 0s and 1s.

## Moore's Law

Moore's Law states that the number of transistors on an integrated chip doubles every 2 years, while the cost and size decrease.

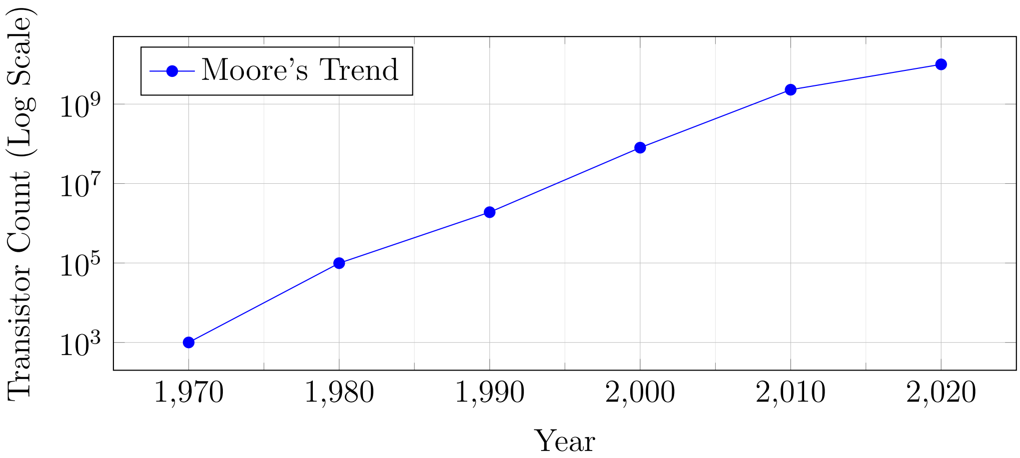



Figure 3: Moore's Law: Exponential Growth of Transistor Density

This trend explains how modern digital devices can pack billions of bits into tiny form factors.

# Understanding Binary Numbers and Arithmetic

## Binary Numbers: MSB and LSB

Binary numbers use only two digits — 0 and 1 — to represent values. Each bit in a binary number corresponds to a power of 2, starting from the right. The rightmost bit is the **Least Significant Bit (LSB)**, and the leftmost bit is the **Most Significant Bit (MSB)**.

**Example**

| | | | | |
|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 0 |

This is the binary number `10110`. Let's break it down using powers of 2:

$$(1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (1 \times 2^1) + (0 \times 2^0) = 16 + 0 + 4 + 2 + 0 = \boxed{22}$$

## Binary to Decimal Conversion

To convert a binary number to decimal:

- Multiply each bit by $2^n$, where $n$ is the position index from right to left (starting at 0).
- Add the results.

**Example**

**Convert 11001 to Decimal**

$$1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 16 + 8 + 0 + 0 + 1 = \boxed{25}$$

## Decimal to Binary Conversion

There are two common methods:

**1. Repeated Division by 2 (Formal Method):**

$$
\begin{aligned}
25 \div 2 &= 12 \text{ remainder } \boxed{1} \\
12 \div 2 &= 6 \text{ remainder } \boxed{0} \\
6 \div 2 &= 3 \text{ remainder } \boxed{0} \\
3 \div 2 &= 1 \text{ remainder } \boxed{1} \\
1 \div 2 &= 0 \text{ remainder } \boxed{1}
\end{aligned}
$$

***Read from bottom to top:*** $\boxed{11001}$

**2. Hit and Trial Method (Largest Power of 2):**

$$25 = 16 + 8 + 1 = 2^4 + 2^3 + 2^0 \Rightarrow \boxed{11001}$$

## Binary Conversion Practice

**Exercise 1**

**Q:** Convert 28 to binary.

**Solution:** Use repeated division by 2:

$$
\begin{aligned}
28 \div 2 &= 14 \text{ remainder } 0 \\
14 \div 2 &= 7 \text{ remainder } 0 \\
7 \div 2 &= 3 \text{ remainder } 1 \\
3 \div 2 &= 1 \text{ remainder } 1 \\
1 \div 2 &= 0 \text{ remainder } 1
\end{aligned}
$$

**Binary (read bottom to top):** $\boxed{11100}$

**Exercise 2**

**Q:** Convert 100101 to decimal.

**Solution:** Multiply each bit with its corresponding power of 2:

$$(1\times 2^5)+(0\times 2^4)+(0\times 2^3)+(1\times 2^2)+(0\times 2^1)+(1\times 2^0) = 32+0+0+4+0+1 = \boxed{37}$$

Exercise 3

**Q:** Convert 39 to binary using both methods.

**Method 1 – Repeated Division:**

$$\begin{aligned}39 \div 2 &= 19 \text{ remainder } 1\\ 19 \div 2 &= 9 \text{ remainder } 1\\ 9 \div 2 &= 4 \text{ remainder } 1\\ 4 \div 2 &= 2 \text{ remainder } 0\\ 2 \div 2 &= 1 \text{ remainder } 0\\ 1 \div 2 &= 0 \text{ remainder } 1\end{aligned}$$

**Binary:** $\boxed{100111}$

**Method 2 – Hit and Trial (Largest Power of 2):**

$$39 = 32 + 4 + 2 + 1 = 2^5 + 2^2 + 2^1 + 2^0 \Rightarrow \boxed{100111}$$

## Binary Addition

**Rules:**

$$\begin{aligned}0 + 0 &= 0\\ 0 + 1 &= 1\\ 1 + 0 &= 1\\ 1 + 1 &= 0 \text{ (carry 1)}\\ 1 + 1 + 1 &= 1 \text{ (carry 1)}\end{aligned}$$

Example

**Q.** Add 11001 and 10110

$$\begin{array}{cccccc} & 1 & 1 & 0 & 0 & 1\\ + & 1 & 0 & 1 & 1 & 0\\ \hline 1 & 0 & 1 & 1 & 1 & 1\end{array}$$

**Result:** $\boxed{101111}$

## Binary Subtraction

**Signed Magnitude Representation:**

- Leftmost bit is the **sign bit** (0 = positive, 1 = negative).
- Remaining bits represent the magnitude.

**Rules for Subtraction:**

$$0 - 0 = 0$$
$$1 - 0 = 1$$
$$1 - 1 = 0$$
$$0 - 1 = 1 \text{ (borrow 1)}$$

**2's Complement Method:** To subtract B from A (A - B), perform:

- Take 2's complement of B
- Add it to A
- Discard carry (if any) for final result

Example

**Q.** Solve 13 – 9 using 2's complement and signed magnitude.

**2's Complement Method:**

$$\begin{aligned} 13 &= \texttt{01101} \\ 9 &= \texttt{01001} \\ \text{2's complement of } 9 &= \texttt{10110 + 1} = \texttt{10111} \end{aligned}$$

$$\begin{array}{cccccc} & 1 & 1 & 0 & 1 & \\ & 0 & 1 & 1 & 0 & 1 \\ + & 1 & 0 & 1 & 1 & 1 \\ \hline 1 & 0 & 0 & 1 & 0 & 0 \end{array}$$

**Discard carry** $\rightarrow$ Result = $\boxed{000100} = \boxed{4}$

**Signed Magnitude Method:**

$$\begin{array}{cccccc} & 0 & 1 & 1 & 0 & 1 \\ - & 0 & 1 & 0 & 0 & 1 \\ \hline & 0 & 0 & 1 & 0 & 0 \end{array}$$

**Result** = $\boxed{4}$

## Subtraction Using Binary Methods

Exercise 4

**Q:** Solve 21 – 15 using both 2's complement and signed magnitude methods.

**2's Complement Method:**

$$\begin{aligned} 21 &= \texttt{10101} \\ 15 &= \texttt{01111} \\ \text{2's complement of 15} &= \texttt{10000 + 1} = \texttt{10001} \end{aligned}$$

$$\begin{array}{cccccc} & 1 & 0 & 1 & 0 & 1 \\ + & 1 & 0 & 0 & 0 & 1 \\ \hline 1 & 0 & 0 & 1 & 1 & 0 \end{array}$$

**Discard carry** → Result = $\boxed{00110}$ = $\boxed{6}$

**Signed Magnitude Method:**

$$\begin{array}{cccccc} \text{Borrow:} & & 1 & & & \\ & \cancel{1}^{0} & \cancel{0} & \cancel{1}^{0} & 0 & 1 \\ - & 0 & 1 & 1 & 1 & 1 \\ \hline & 0 & 0 & 1 & 1 & 0 \end{array}$$

**Result:** $\boxed{6}$

Exercise 5

**Q:** Solve 15 – 9 using both 2's complement and signed magnitude methods.

**2's Complement Method:**

$$\begin{aligned} 15 &= \texttt{01111} \\ 9 &= \texttt{01001} \\ \text{2's complement of 9} &= \texttt{10110 + 1} = \texttt{10111} \end{aligned}$$

$$\begin{array}{cccccc} & 0 & 1 & 1 & 1 & 1 \\ + & 1 & 0 & 1 & 1 & 1 \\ \hline 1 & 0 & 0 & 1 & 1 & 0 \end{array}$$

**Discard carry** $\rightarrow$ Result $= \boxed{00110} = \boxed{6}$

**Signed Magnitude Method:**

$$\begin{array}{cccccc} & 0 & 1 & 1 & 1 & 1 \\ - & 0 & 1 & 0 & 0 & 1 \\ \hline & 0 & 0 & 1 & 1 & 0 \end{array}$$

**Result** $= \boxed{6}$

### Exercise 6

**Q:** Solve 5 – 12 using 2's complement method.

**2's Complement Method:**

$$\begin{aligned} 5 &= \texttt{00101} \\ 12 &= \texttt{01100} \\ \text{2's complement of } 12 &= \texttt{10011 + 1} = \texttt{10100} \end{aligned}$$

$$\begin{array}{cccccc} & 0 & 0 & 1 & 0 & 1 \\ + & 1 & 0 & 1 & 0 & 0 \\ \hline & 1 & 1 & 0 & 0 & 1 \end{array}$$

MSB is 1 $\rightarrow$ Negative. Take 2's complement of `1001`:

$$\text{1's comp: } \texttt{0110}, \quad \text{add 1: } \texttt{0111} = \boxed{7}$$

**Result** $= \boxed{-7}$

Exercise 7

**Q:** Solve –15 – 21 using 2's complement method.

**2's Complement Method:**

$$\begin{aligned} -15 &= \text{2's comp of } \texttt{0001111} = \texttt{1110000 + 1} = \texttt{1110001} \\ -21 &= \text{2's comp of } \texttt{0010101} = \texttt{1101010 + 1} = \texttt{1101011} \end{aligned}$$

$$\begin{array}{rccccccc} \text{Carry:} & 1 & & & & 1 & 1 & \\ & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ + & 1 & 1 & 0 & 1 & 0 & 1 & 1 \\ \hline \cancel{1} & 1 & 0 & 1 & 1 & 1 & 0 & 0 \end{array}$$

**Interpretation:**

- Result = `1011100` → MSB = 1 → Negative
- 2's comp of `1011100` = `0100100` = $\boxed{36}$

**Result:** $\boxed{-36}$

**Why did we add extra 0's to the MSB?**

In 2's complement, we must ensure that all operands have the same bit width before performing binary arithmetic. This is essential for correctness and proper interpretation of signed numbers.

- The values $-15$ and $-21$ exceed the representable range of 5-bit 2's complement, which is $[-16, +15]$.
- Therefore, we extend the system to 6 or 7 bits, which supports a wider range.
- A leading `0` is added to the MSB (for positive values) or a `1` (for negative values) during sign extension to preserve the correct value and sign.

This guarantees that all binary values are correctly aligned and interpreted, and the final result of the addition remains valid in the chosen bit system.

# Contribution Statement

1. **Tarunika Ravikumar:** Structuring and formatting of lecture notes on Overleaf.
2. **Sunaina Reji Baker:** Taking detailed notes for lecture 1.
3. **Shama Majed Alshehhi:** Taking detailed notes for lecture 2.
4. **Noura Mohsin Alameri:** Taking detailed notes for lecture 3.